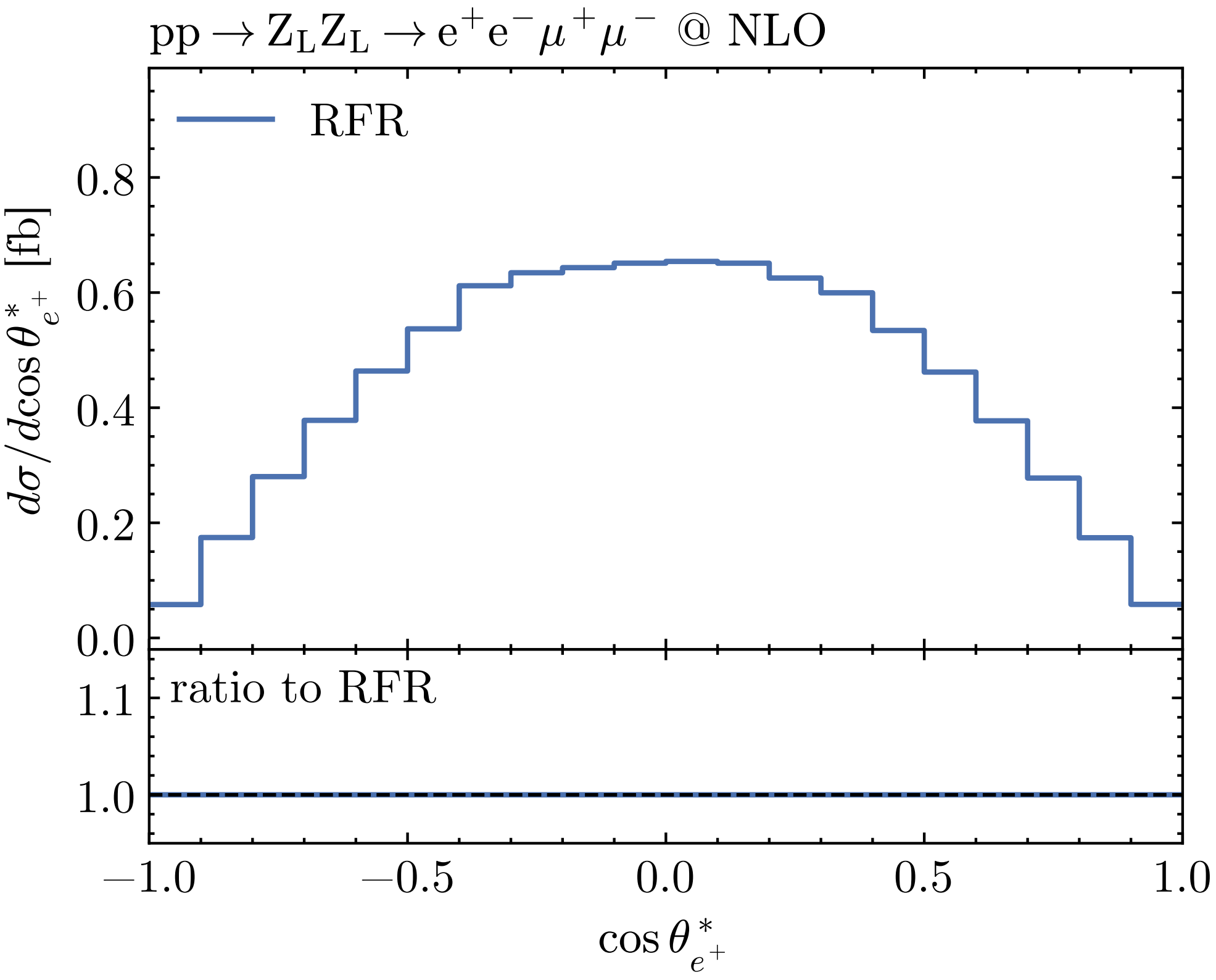
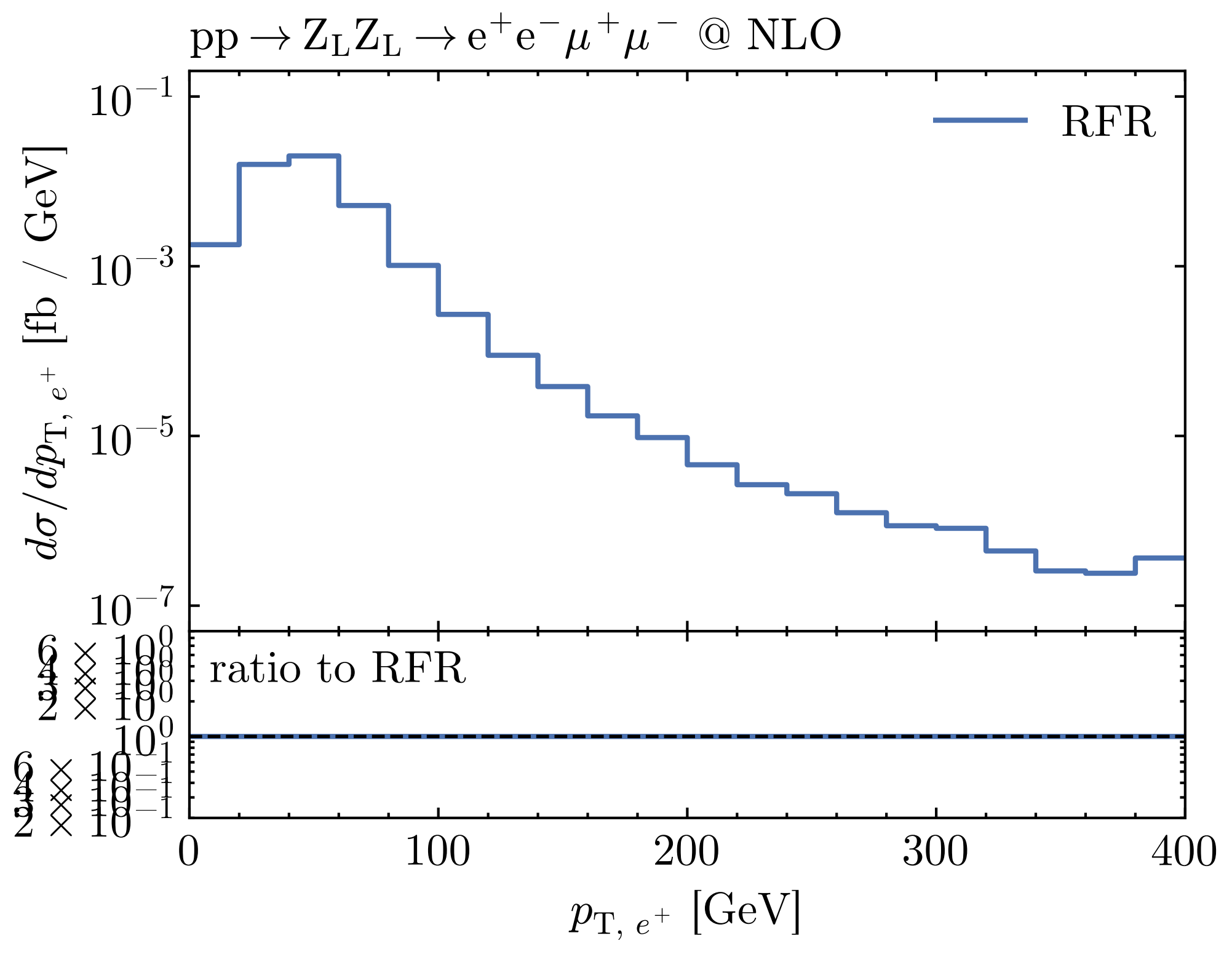
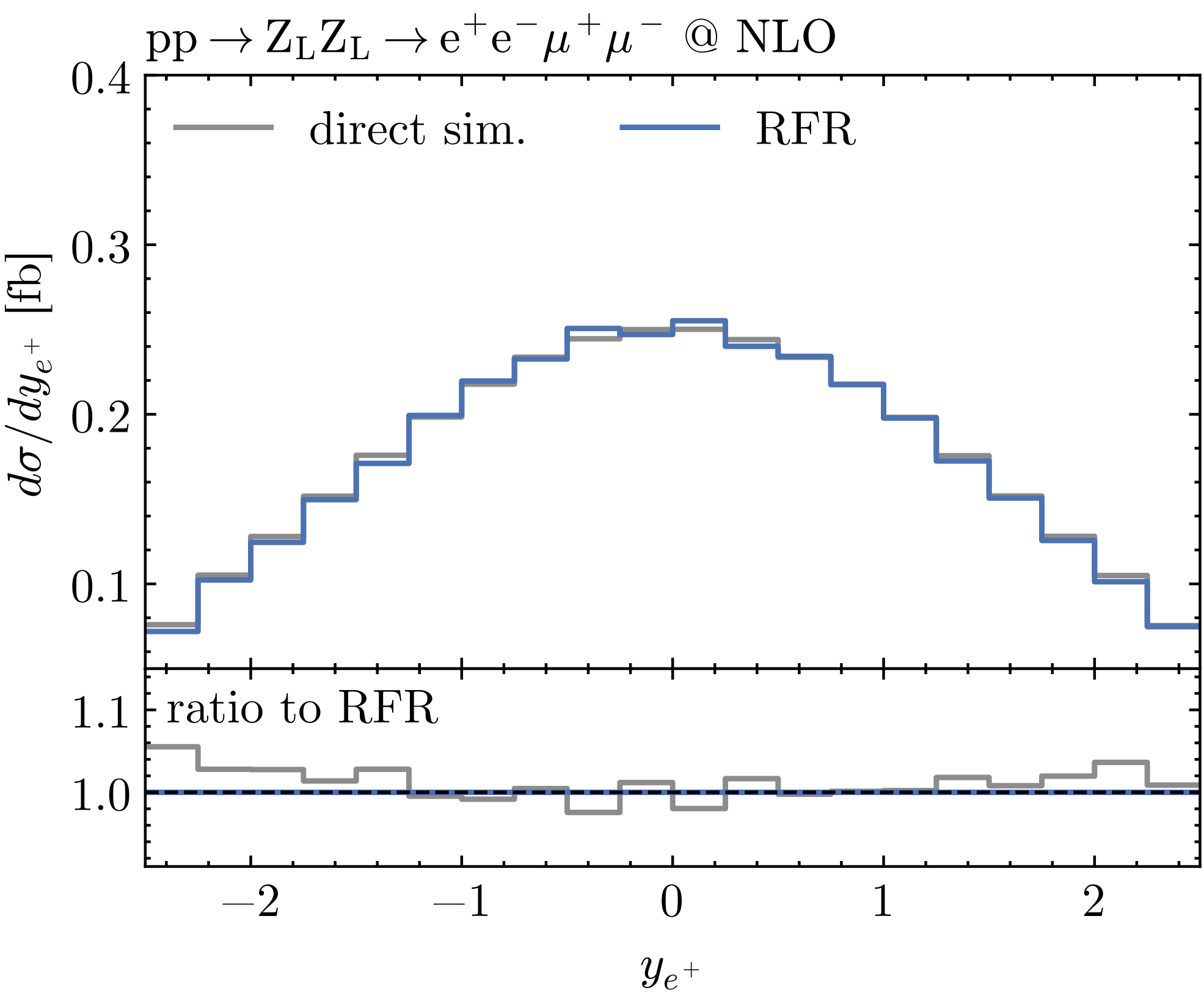
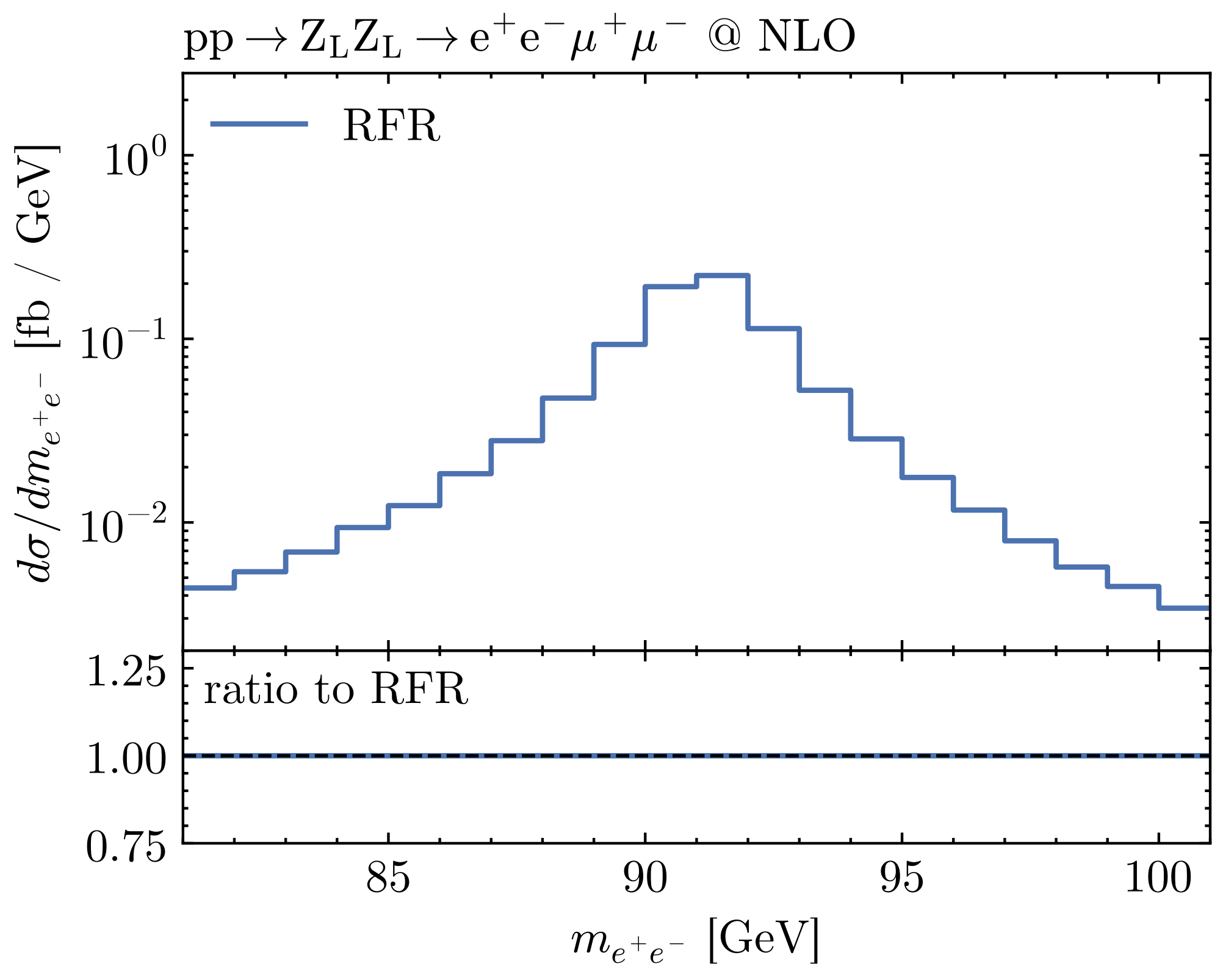
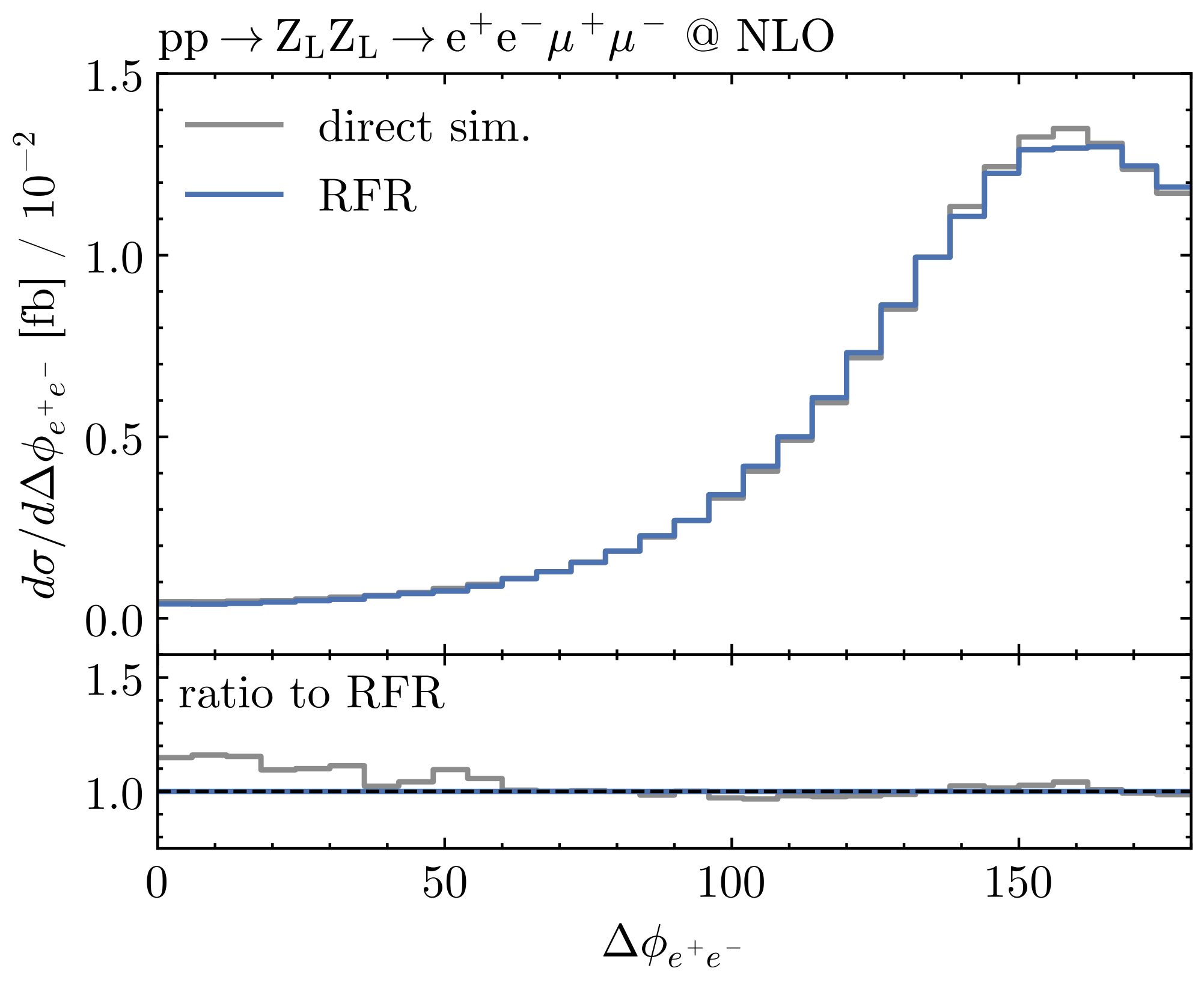
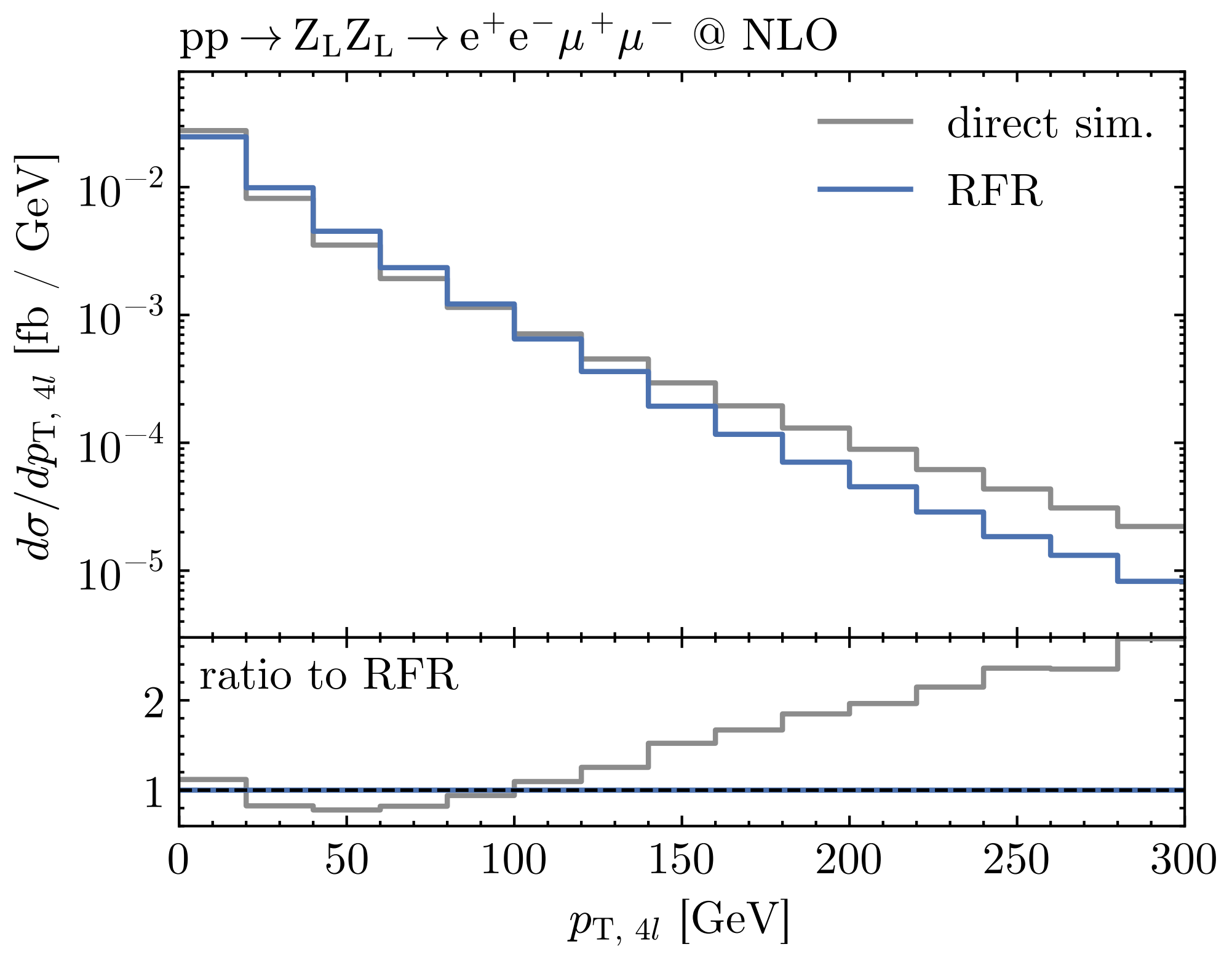

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO
RFR
$d\sigma/d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to RFR
1.0
1.1
−1.0
−0.5
0.0
0.5
1.0
$\cos\theta^*_{e^+}$

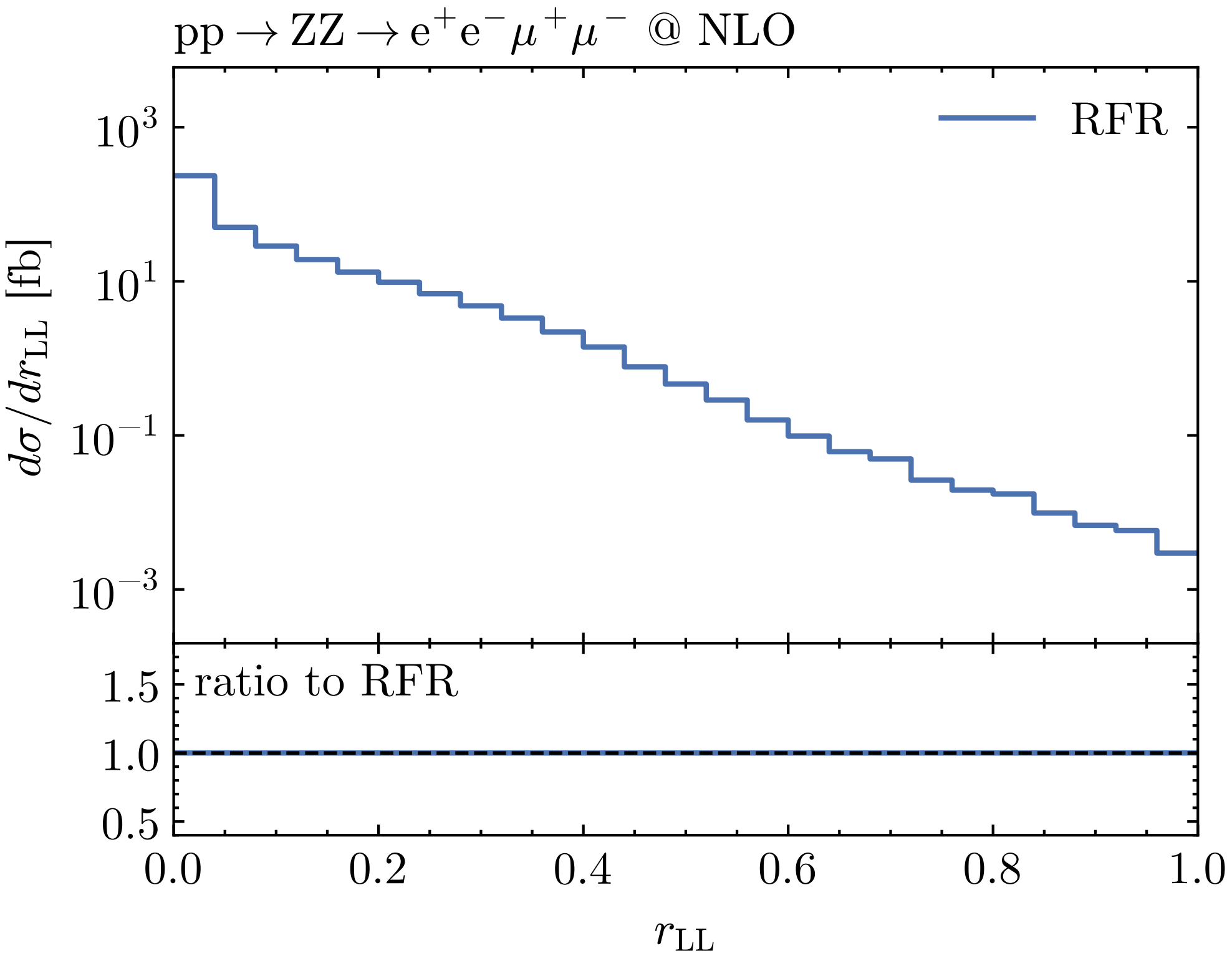

pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻ @ NLO
RFR
dσ/dp_T, e⁺ [fb / GeV]
ratio to RFR
p_T, e⁺ [GeV]
0
100
200
300
400


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO
direct sim.
RFR
$d\sigma/dy_{e^+}$ [fb]
0.4
0.3
0.2
0.1
ratio to RFR
1.1
1.0
−2
−1
0
1
2
y_{e^+}


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO
RFR
10^0
10^{-1}
10^{-2}
$d\sigma/dm_{e^+e^-}$ [fb / GeV]
1.25
1.00
0.75
ratio to RFR
85
90
95
100
$m_{e^+e^-}$ [GeV]


pp → Z_L Z_L → e+ e− μ+ μ− @ NLO
direct sim.
RFR
dσ/dΔφ_{e+e−} [fb] / 10^{−2}
1.5
1.0
0.5
0.0
ratio to RFR
1.5
1.0
0
50
100
150
Δφ_{e+e−}


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO
direct sim.
RFR
$d\sigma/dp_{\mathrm{T},\,4l}$ [fb / GeV]
10^{-2}
10^{-3}
10^{-4}
10^{-5}
ratio to RFR
2
1
0
50
100
150
200
250
300
$p_{\mathrm{T},\,4l}$ [GeV]


$\mathrm{pp} \to \mathrm{ZZ} \to \mathrm{e}^+\mathrm{e}^-\mu^+\mu^-$ @ NLO
RFR
10^3
10^1
10^{-1}
10^{-3}
$d\sigma/dr_{\mathrm{LL}}$ [fb]
ratio to RFR
1.5
1.0
0.5
0.0
0.2
0.4
0.6
0.8
1.0
r_{LL}